

CPE301 – SPRING 2024

Design Assignment 4

Student Name: Joshua Martinez

Student #: 5007531628

Student Email: marti87@unlv.nevada.edu

Primary Github address: <https://github.com/JoshuaMa2003>

Directory: https://github.com/JoshuaMa2003/submission_da

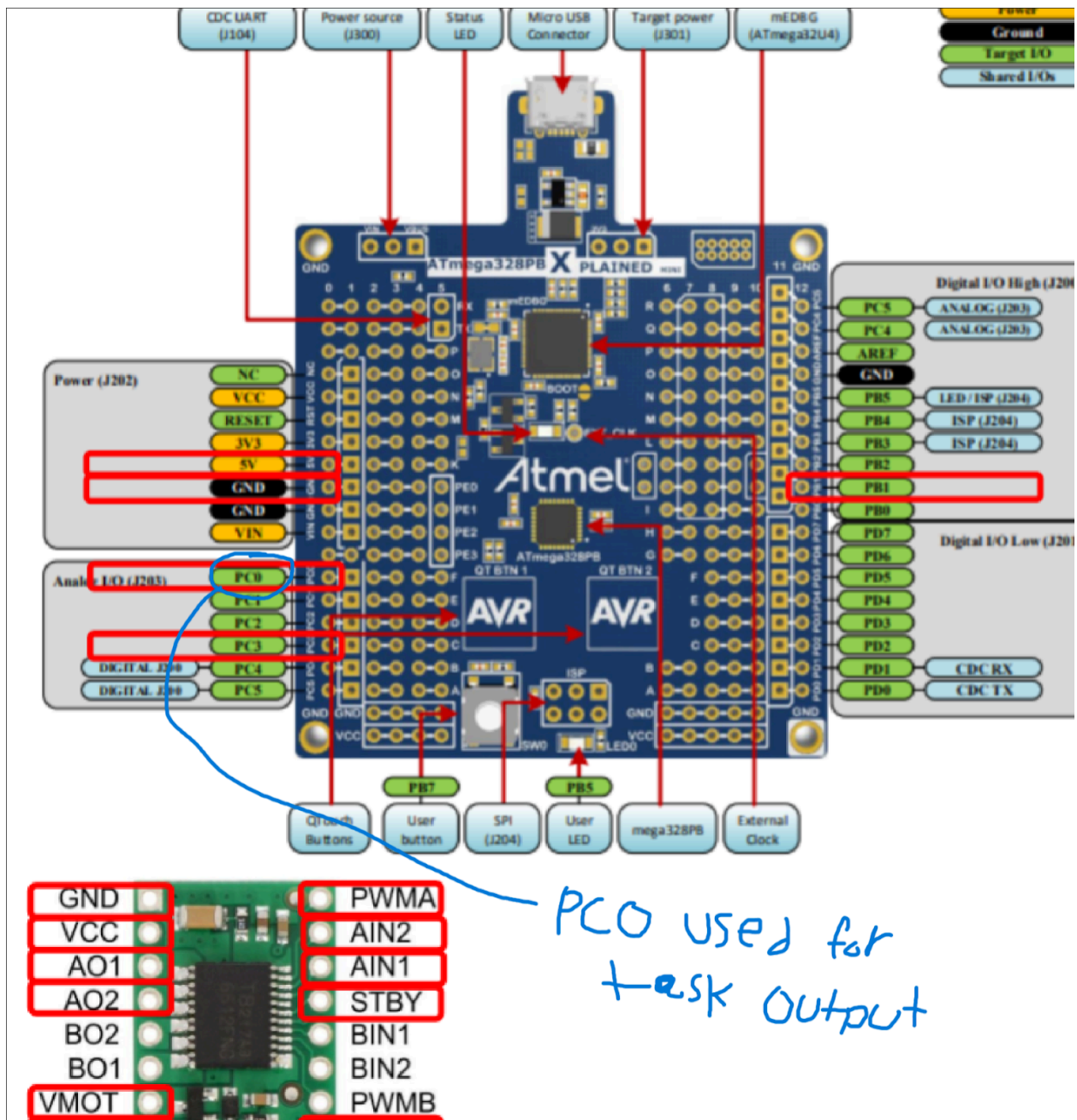
Video Playlist:

https://www.youtube.com/playlist?list=PLZ09MA_uFt63me60r5XWtDUj2m21n4nUP

1. COMPONENTS LIST AND DIAGRAM SHOWING PINS USED

List of Components used

- Microchip Studio Debugger
- Microchip Studio Simulator
- ATmega328PB Microcontroller
- ATmega328PB Microcontroller Potentiometer



2. INITIAL CODE OF TASK

```
/*
 * CPE301_DesignAssignment4_C.c
 *
 * Created: 4/9/2024 8:35:41 AM
 * Author : joshu
 */

#define F_CPU 16000000UL //define cpu clock frequency
#define BAUDRATE 9600 //define baud rate for usart
#define BAUD_PRESCALE (((F_CPU / (BAUDRATE * 16UL))) - 1) //define baud rate prescaler value

//include needed avr headers
#include <avr/io.h>
#include <stdlib.h>
#include <string.h>
#include <util/delay.h>
#include <avr/interrupt.h>

char out_str[51]; //define buffer for output string

int adc_init(void)
{
    ADMUX |= (1 << REFS0); //reference voltage for adc
    ADCSRA = (1 << ADEN) | (1 << ADSC) | (1 << ADIF) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0); //adc configuration
    ADCSRB = (1 << ADTS2) | (1 << ADTS1); //adc trigger source

    while(ADCSRA & (1 << ADSC)); //wait for adc
    return ADC; //return adc value
}

void timer_init(void)
{
    TCNT1 = 65380; //preload timer1 for delay adjustments
    TCCR1B |= (1 << CS11); //set prescaler to 8
    TIMSK1 = (1 << TOIE1); //enable timer1 overflow interrupt
    sei(); //enable global interrupt
}

void usart_init(void)
{
    UBRR0H = (uint8_t)(BAUD_PRESCALE >> 8); // set baud rate high
    UBRR0L = (uint8_t)(BAUD_PRESCALE); //set baud rate low
    UCSRB = (1 << RXEN0) | (1 << TXEN0); //enable receiver and transmitter
    UCSRC = (1 << UCSZ01) | (1 << UCSZ00); //set frame format
}

int main(void)
{
    memset(out_str, ' ', 49); //initialize out_str with spaces
    out_str[50] = '\0'; //null terminate string

    timer_init(); //timer initialization
    usart_init(); //usart initialization
    adc_init(); //adc initialization

    while(1)
    {
        //infinite loop
    }
    return 0;
}

void usart_send(unsigned char ch)
{
    while(!(UCSR0A & (1 << UDRE0))); //wait for data register to be empty
    UDR0 = ch; //send character
}

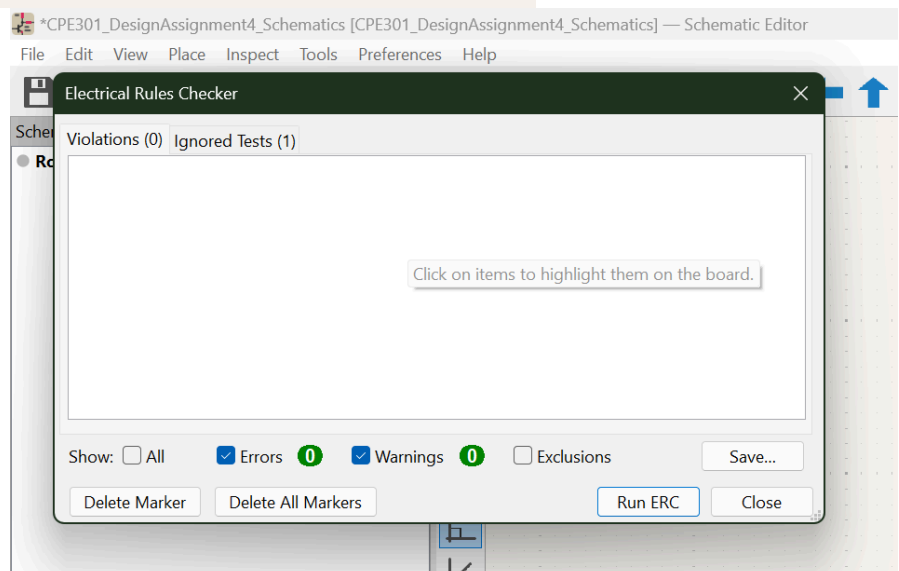
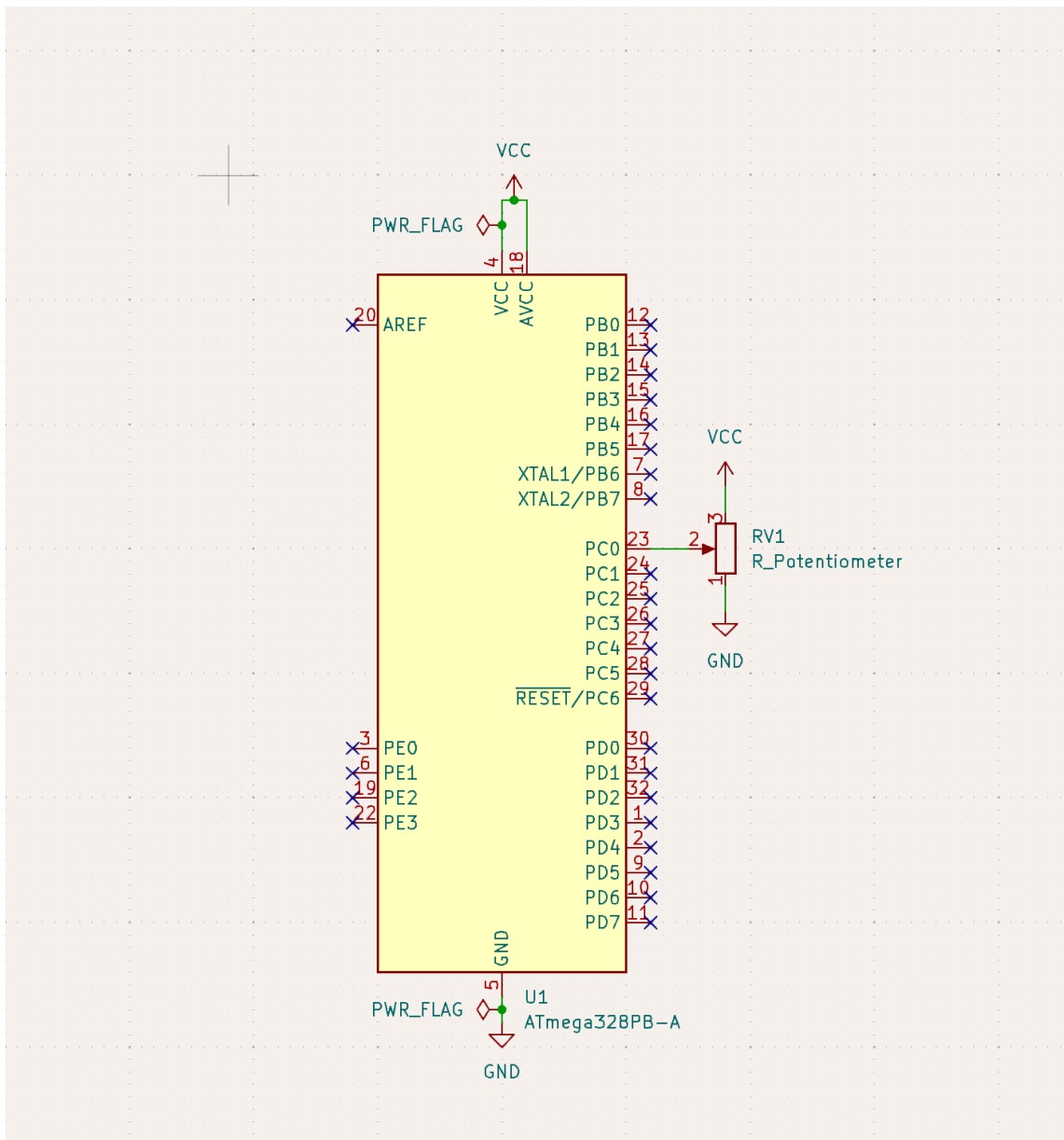
//print a string over usart
void usart_print(char* str)
{
    int i = 0;
    while(str[i] != '\0')
    {
        usart_send(str[i]);
        i++;
    }
}

//send string over to usart
void usart_putstr(char* StringPtr)
{
    while((*StringPtr) != '\0')
    {
        while(!(UCSR0A & (1 << UDRE0)));
        UDR0 = *StringPtr;
        StringPtr++;
    }
}

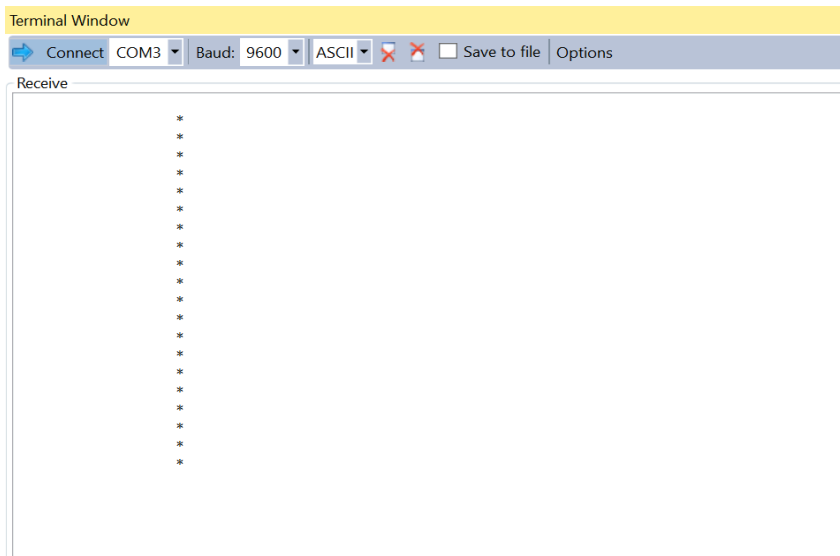
ISR(TIM1_OVF_vect)
{
    ADCSRA |= (1 << ADSC); //start adc conversion
    while((ADCSRA & (1 << ADIF)) == 0); //wait for conversion to complete
    ADCSRA |= (1 << ADIF); //clear adc interrupt flag

    int adcVal = ADCL; //read low byte
    adcVal = adcVal | (ADCH << 8); //read high byte, combine with low byte
    int8_t index = adcVal/20.48; //convert adc value to index
    out_str[index] = '*'; //mark current position in buffer
    usart_putstr(out_str); //send buffer over to usart
    usart_putstr("\n"); //send newline to usart
    out_str[index] = ' '; //clear marked position in buffer
    TCNT1 = 65380; //reload timer
}
```

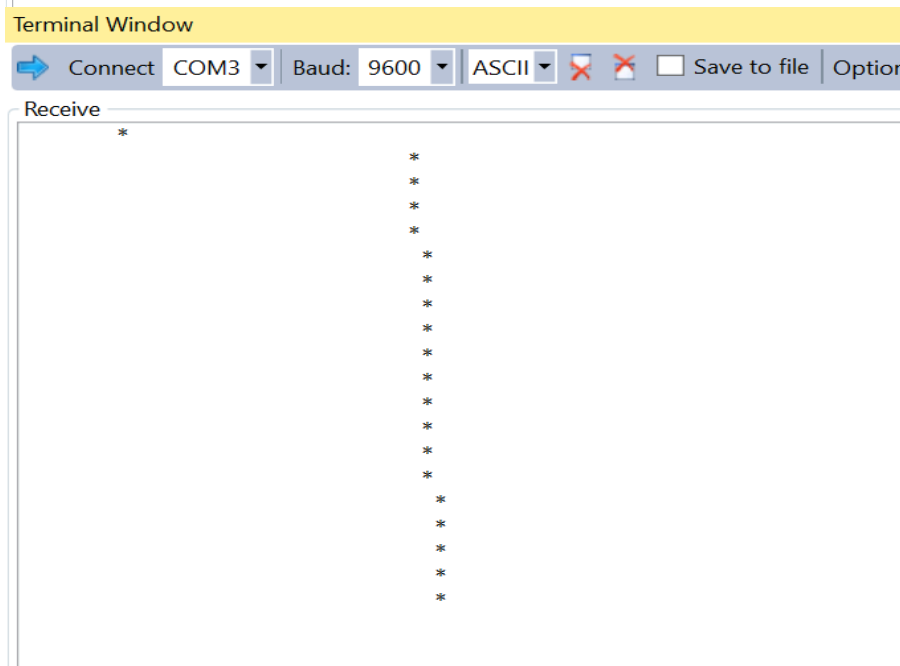
3. KICAD SCHEMATICS FOR TASK



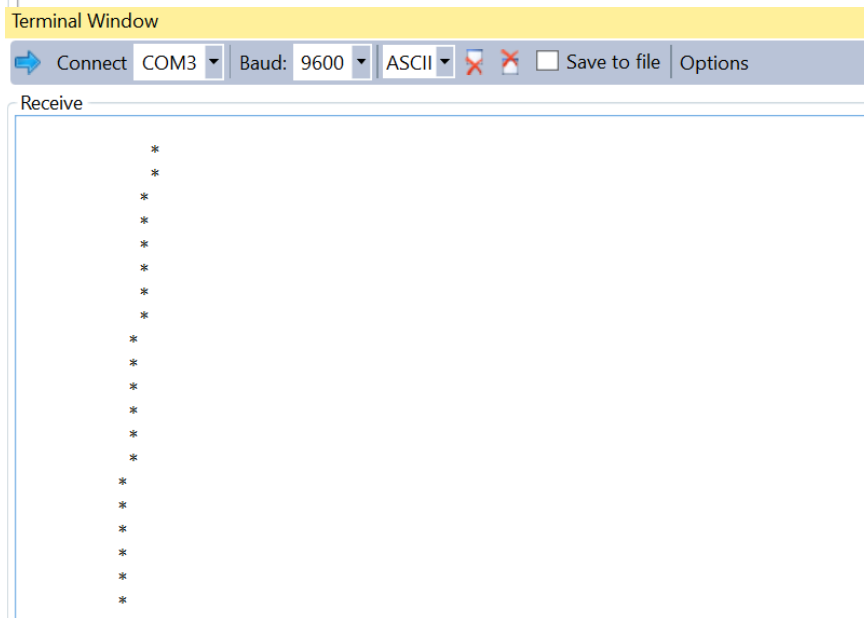
4. SCREENSHOTS OF EACH TASK OUTPUT



Steady State

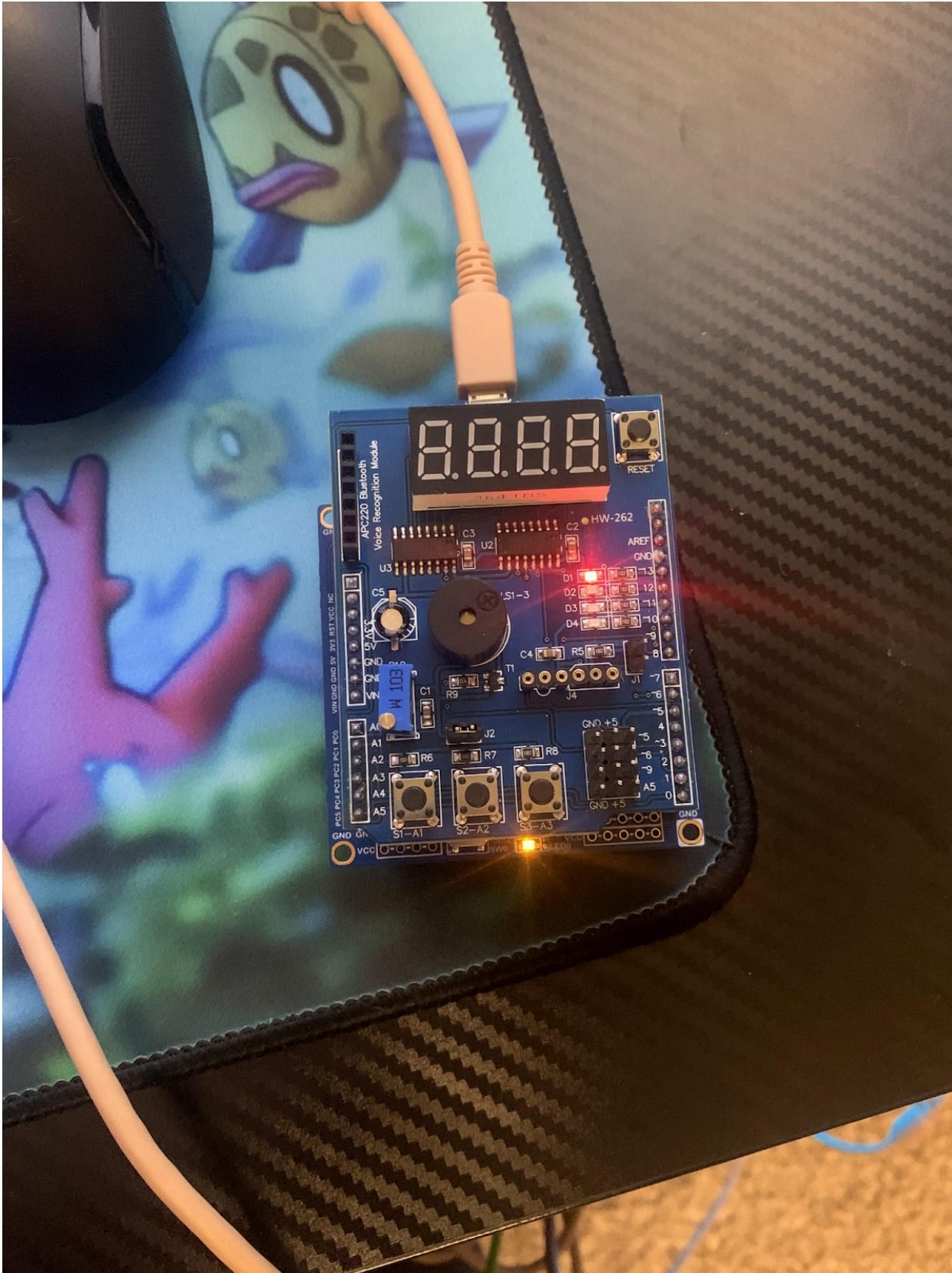


Increasing Voltage



Decreasing Voltage

5. SCREENSHOTS OF EACH DEMO



6. VIDEO LINKS OF EACH DEMO

Task Demo: <https://youtu.be/DKMexjvHnAA?feature=shared>

Student Academic Misconduct Policy

<http://studentconduct.unlv.edu/misconduct/policy.html>

"This assignment submission is my own, original work".

JOSHUA MARTINEZ